a) P lies within exactly one of the leaves' rectangles, R: then just add P to R, and move to next step.

b) P lies within the overlapping region between two or more leaves' bounding rectangles. For example, referring to figure 10.6, it might lie in the intersection of R12 and R14. In this case, we need to decide where to add P: the heuristic used to make these decisions will determine the shape of the tree (as an example, one heuristic could be just adding it to the rectangle with fewer elements).

c) If P lies outside of all rectangles at the leaves level, then we need to find the closest leaf L and add P to it (again, we can use more complex heuristics than just the Euclidean distance to decide).

2. Add the points to leaf's rectangle R, and check how many points it contains afterwards:

a) if, after the new point is added, the still has at most M points, then we are done.

b) otherwise, we need to split R into two new rectangles: R1 and R2, and go to step 3.

3. Remove R from its parent RP and add R1 and R2 to RP. If RP now has more than M children, split it and repeat this step recursively.

a) If R was the root, we obviously can't remove it from its parent: we just create a new root and set R1 and R2 as children.

To complete the insertion algorithm outlined above, we need to provide a few heuristics: those to break ties for overlapping rectangles and to choose the closest rectangle,